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Side sill reinforcement structure for electric vehicle

Abstract

A side sill reinforcement structure includes a battery unit for drive, a side sill extending in a vehicle front-rear direction on an outer side in a vehicle width direction of the battery unit, and a reinforcing member that connects the battery unit and the side sill. The reinforcing member includes a closed cross-sectional portion group having a first closed cross-sectional portion disposed on the innermost side in the vehicle width direction and a second closed cross-sectional portion disposed adjacent to the first closed cross-sectional portion. The first closed cross-sectional portion includes a first upper wall and a first lower wall. The second closed cross-sectional portion includes a second upper wall and a second lower wall. The thickness of the first upper wall is larger than the thickness of the second upper wall, and the thickness of the first lower wall is larger than the thickness of the second lower wall.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a national phase application in the United States of International Patent Application No. PCT/JP2021/036637 with an international filing date of Oct. 4, 2021, which claims priority of Japanese Patent Application No. 2020-198777 filed on Nov. 30, 2020, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a side sill reinforcement structure for an electric vehicle.

BACKGROUND ART

(3) Electric vehicles are required to have higher collision safety performance than general fuel driven vehicles from the viewpoint of battery protection. In an electric vehicle, in order to secure a cruising range, a battery is widely arranged on an entire underfloor surface of a vehicle interior in many cases, and thus high collision safety performance at the time of side collision (hereinafter, also referred to as side collision) is particularly required. That is, when an object such as a pole collides with a vehicle body side portion due to spinning of the vehicle body or the like, it is necessary to protect the battery and the vehicle interior without being damaged.

(4) For example, JP 2015-74244 A and JP 2013-256265 A disclose a vehicle body lower structure capable of suppressing deformation of a vehicle interior and improving side collision performance of a vehicle body. In the vehicle body lower structure, in order to obtain high side collision performance, a reinforcing member is disposed under a columnar member on a vehicle body lower side portion referred to as a side sill to improve the side collision performance.

SUMMARY OF THE INVENTION

(5) The structures of JP 2015-74244 A and JP 2013-256265 A are structures in which an extruded shape material reinforcing material is disposed under a side sill, and are structures intended to absorb energy at the time of side collision by using these. In these structures, the cross-sectional wall thickness of the extruded shape material constituting the reinforcing material is set to be the same in each part. However, in order to achieve both securing battery protection performance and reducing the weight of components, it is desirable to optimize the cross-sectional wall thickness of each part by utilizing the advantage of the extruded shape material. However, such a structure that take these into account are not found.

(6) An object of the present invention is to efficiently improve battery protection performance in a side sill reinforcement structure for an electric vehicle.

(7) The present invention provides a side sill reinforcement structure for an electric vehicle including: a battery unit for drive disposed in a vehicle interior lower portion of an electric vehicle; a side sill extending in a vehicle front-rear direction on an outer side in a vehicle width direction of the battery unit; and a reinforcing member configured to connect the battery unit and the side sill. The reinforcing member is an extruded material made of an aluminum alloy, and includes a closed cross-sectional portion group in which a plurality of closed cross-sectional portions are continuous in the vehicle width direction in a cross section perpendicular to the vehicle front-rear direction. The closed cross-sectional portion group includes a first closed cross-sectional portion disposed on an innermost side in the vehicle width direction and a second closed cross-sectional portion disposed adjacent to the first closed cross-sectional portion. The first closed cross-sectional portion includes a first upper wall and a first lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in a vehicle vertical direction. The second closed cross-sectional portion includes a second upper wall and a second lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction. A thickness of the first upper wall is larger than a thickness of the second upper wall, and a thickness of the first lower wall is larger than a thickness of the second lower wall.

(8) According to this configuration, since the extruded material made of an aluminum alloy is adopted as the reinforcing member, the thickness can be partially changed. Accordingly, the thicknesses of the first upper wall and the first lower wall can be made larger than the thicknesses of the second upper wall and the second lower wall, respectively. In particular, since being disposed on the innermost side in the vehicle width direction in the closed cross-sectional portion group, the first closed cross-sectional portion is a closed cross-sectional portion closest to the battery unit. Furthermore, since extending in the vehicle width direction, the first upper wall and the first lower wall of the first closed cross-sectional portion are portions that receive a side collision load. Therefore, increasing the thickness of the portions efficiently can improve battery protection performance while suppressing an increase in weight and an increase in cost as compared with a case of adopting a reinforcing member having a uniform thickness. Here, the electric vehicle widely refers to a vehicle having a drive battery on the lower side of the vehicle interior. In addition, the battery unit is a unit including a battery and a battery case.

(9) A thickness of the first upper wall may be 1.2 times or more a thickness of the second upper wall, and a thickness of the first lower wall may be 1.2 times or more a thickness of the second lower wall.

(10) According to this configuration, it is possible to secure sufficient battery protection performance by specifically defining a necessary thickness.

(11) The first closed cross-sectional portion may be joined to the battery unit by a bolt.

(12) According to this configuration, since the first closed cross-sectional portion having a relatively large thickness is firmly joined to the battery unit by the bolt, high coupling rigidity can be secured.

(13) The closed cross-sectional portion group may include a third closed cross-sectional portion disposed on an outermost side in the vehicle width direction and a fourth closed cross-sectional portion disposed on an inner side in the vehicle width direction adjacent to the third closed cross-sectional portion. The third closed cross-sectional portion may include a third upper wall and a third lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction. The fourth closed cross-sectional portion may include a fourth upper wall and a fourth lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction. A thickness of the third upper wall may be larger than a thickness of the fourth upper wall, and a thickness of the third lower wall may be larger than a thickness of the fourth lower wall.

(14) According to this configuration, by improving the strength of the outermost portion (third closed cross-sectional portion) in the vehicle width direction first receiving deformation in the side collision, the peak value of the deformation load received at the initial stage of the side collision can be increased. When the peak value of the deformation load received at the initial stage of deformation is increased as described above, the load variation according to the progress of deformation can be suppressed, and the energy absorption efficiency can be improved.

(15) A thickness of the first upper wall may be smaller than a thickness of the third upper wall, and a thickness of the first lower wall may be smaller than a thickness of the third lower wall.

(16) According to this configuration, the first upper wall and the first lower wall do not become excessively thick, so that it is possible to suppress an excessive increase in weight. In addition, in the extruded material made of an aluminum alloy, when a significant difference is provided in thickness, there is a possibility that extrusion processing cannot be stably executed. Therefore, by providing the upper limit to the thickness as described above, the extrusion processing can be stably executed.

(17) The third closed cross-sectional portion may be joined to the side sill by a bolt.

(18) According to this configuration, since the third closed cross-sectional portion having a relatively large thickness is firmly joined to the side sill by the bolt, high coupling rigidity can be secured.

(19) Each closed cross-sectional portion included in the closed cross-sectional portion group may include an upper wall and a lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction.

Thicknesses of the upper wall and the lower wall may be sequentially decreased from an outer side to an inner side in the vehicle width direction excluding the first closed cross-sectional portion.

(20) According to this configuration, since the deformation strength of the reinforcing member is sequentially lowered from the outer side to the inner side in the vehicle width direction in accordance with the deformation region being expanded in the vehicle width direction according to the progress of the side collision, the load according to the progress of the deformation can be further equalized. Therefore, energy absorption efficiency can be further improved. In addition, since the application is excluded with respect to the first closed cross-sectional portion, the first closed cross-sectional portion can be configured less likely to be deformed even when the region on the outer side in the vehicle width direction from the first closed cross-sectional portion is crushed. Therefore, higher battery protection performance can be secured.

(21) The reinforcing member may include a lower side extending portion extending downwards the battery unit. A cross member extending in the vehicle width direction may be provided under the battery unit. The lower side extending portion may be joined to the cross member.

(22) According to this configuration, when a side collision load is applied to the first closed cross-sectional portion, the load can be transmitted to the cross member through the lower side extending portion. Therefore, higher battery protection performance can be secured.

(23) According to the present invention, battery protection performance can be efficiently improved in a side sill reinforcement structure for an electric vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of an electric vehicle;
- (2) FIG. 2 is a cross-sectional view of a side sill reinforcement structure for an electric vehicle according to a first embodiment of the present invention;
- (3) FIG. 3 is a cross-sectional view of a reinforcing member in FIG. 2;
- (4) FIG. 4 is a cross-sectional view of a reinforcing member of a side sill reinforcement structure for an electric vehicle according to a second embodiment;
- (5) FIG. 5 is a cross-sectional view of a side sill reinforcement structure for an electric vehicle according to a third embodiment;
- (6) FIG. 6 is a cross-sectional view of a side sill reinforcement structure for an electric vehicle according to a first modification of the third embodiment;
- (7) FIG. 7 is a cross-sectional view of a side sill reinforcement structure for an electric vehicle according to a second modification of the third embodiment;
- (8) FIG. 8 is a cross-sectional view of a side sill reinforcement structure for an electric vehicle according to a fourth embodiment; and
- (9) FIG. 9 is a cross-sectional view of the reinforcing member in FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

- (10) Hereinafter, embodiments of the present invention will be described with reference to the accompanying drawings.

First Embodiment

- (11) Referring to FIG. 1, an electric vehicle **1** is a vehicle that travels by driving a motor (not shown) by electric power supplied from a battery **11**. The electric vehicle **1** widely refers to a vehicle including a drive battery on the underside of the vehicle interior R, and can be, for example, an electric automobile, a plug-in hybrid vehicle, or the like. The type of the vehicle is not particularly limited, and may be a passenger car, a truck, a maintenance vehicle, other mobility, or the like. Hereinafter, as the electric vehicle **1**, a side sill reinforcement structure for an electric automobile of the present embodiment will be described by taking a case of a passenger car type electric vehicle as shown in FIG. 1 as an example.
- (12) In FIG. 1, the front-rear direction of the electric vehicle **1** is indicated by an X direction, and the vertical direction is indicated by a Z direction. The same notation also applies to the subsequent figures, and the vehicle width direction is indicated by a Y direction in FIG. 2 and subsequent figures.
- (13) In the electric vehicle **1**, the battery unit **10** is mounted on the substantially entire surface under the floor of the vehicle interior R at the central portion of the vehicle body. The battery unit **10** includes a drive battery **11** and a battery case **12** housing the battery **11**.
- (14) Referring to FIG. 2, the battery case **12** includes an upper case **13** and a lower case **14**. The upper case **13** includes a housing portion **13a** having an upward-recessed shape and a flange portion **13b** provided around the housing portion **13a** in a plan view. The lower case **14** includes a housing portion **14a** having a downward-recessed shape and a flange portion **14b** provided around the housing portion **14a** in a plan view. The upper case **13** and the lower case **14** are joined by bonding the flange portions **13b** and **14b**. The battery **11** is housed in a housing space S1 defined by the housing portion **13a** of the upper case **13** and the housing portion **14a** of the lower case **14**.
- (15) A hollow side sill **20** extending in the vehicle front-rear direction below the vehicle interior R is disposed on the outer side (left side in FIG. 2) in the vehicle width direction of the battery unit

10. The side sill **20** is a strength member constituting an outer side surface of a lower portion of the electric vehicle **1**.

(16) In the side sill reinforcement structure for an electric vehicle of the present embodiment, the side sill **20** includes an outer member **21** disposed on the outer side in the vehicle width direction and an inner member **22** disposed on the inner side in the vehicle width direction. Both the outer member **21** and the inner member **22** are formed by bending a sheet metal into a hat shape, and are bonded so as to form an internal space **S2**. The outer member **21** and the inner member **22** may be made of, for example, steel. Alternatively, the outer member **21** and the inner member **22** may be made of, for example, an aluminum alloy.

(17) A floor panel **30** is disposed above the battery unit **10**. The floor panel **30** is a plate material constituting a lower surface of the vehicle interior **R**. A cross member **31** extending in the vehicle width direction is disposed above the floor panel **30**. Although not shown in detail, a plurality of cross members **31** are provided at intervals in the vehicle front-rear direction. The upper portions of a pair of side sills **20** provided on both outer sides of the vehicle interior **R** in the vehicle width direction are connected to each other by the plurality of cross members **31**. It should be noted that in FIG. 2, only one of the pair of side sills **20** is shown.

(18) A reinforcing member **40** made of an extruded material made of an aluminum alloy is disposed below the side sill **20**. The reinforcing member **40** connects the battery unit **10** and the side sill **20**.

(19) The reinforcing member **40** extends in the vehicle front-rear direction along the side sill **20**. The reinforcing member **40** has a closed cross-sectional portion group **40A** in which a plurality of closed cross-sectional portions are continuous in the vehicle width direction in a cross section perpendicular to the vehicle front-rear direction.

(20) In the present embodiment, the closed cross-sectional portion group **40A** includes a first closed cross-sectional portion **41** disposed on the innermost side in the vehicle width direction (right side in FIG. 2) and a second closed cross-sectional portion **42** disposed adjacent to the first closed cross-sectional portion **41**. In addition, the closed cross-sectional portion group **40A** includes a third closed cross-sectional portion **43** disposed on the outermost side in the vehicle width direction (left side in FIG. 2) and a fourth closed cross-sectional portion **44** disposed adjacent to the third closed cross-sectional portion **43**. In addition, the closed cross-sectional portion group **40A** includes a fifth closed cross-sectional portion **45** disposed between the second closed cross-sectional portion **42** and the fourth closed cross-sectional portion **44**. That is, the closed cross-sectional portion group **40A** includes five closed cross-sectional portions **41** to **45** continuous in the vehicle width direction. The five closed cross-sectional portions **41** to **45** have rectangular shapes having substantially the same size.

(21) Referring to FIG. 3, the first closed cross-sectional portion **41** includes a first upper wall **41a** and a first lower wall **41b** that extend in the vehicle width direction and are disposed to face each other while being separated from each other in the vehicle vertical direction. The first upper wall **41a** is disposed on a relatively upper side, and the first lower wall **41b** is disposed on a relatively lower side. The first upper wall **41a** and the first lower wall **41b** have the same thickness. However, the thicknesses of the first upper wall **41a** and the first lower wall **41b** may be different.

(22) The second closed cross-sectional portion **42** includes a second upper wall **42a** and a second lower wall **42b** that extend in the vehicle width direction and are disposed to face each other while being separated from each other in the vehicle vertical direction. The second upper wall **42a** is disposed on a relatively upper side, and the second lower wall **42b** is disposed on a relatively lower side. The second upper wall **42a** and the second lower wall **42b** have the same thickness. However, the thicknesses of the second upper wall **42a** and the second lower wall **42b** may be different.

(23) The third closed cross-sectional portion **43** includes a third upper wall **43a** and a third lower wall **43b** that extend in the vehicle width direction and are disposed to face each other while being separated from each other in the vehicle vertical direction. The third upper wall **43a** is disposed on a relatively upper side, and the third lower wall **43b** is disposed on a relatively lower side. The third

upper wall **43a** and the third lower wall **43b** have the same thickness. However, the thicknesses of the third upper wall **43a** and the third lower wall **43b** may be different.

(24) The fourth closed cross-sectional portion **44** includes a fourth upper wall **44a** and a fourth lower wall **44b** that extend in the vehicle width direction and are disposed to face each other while being separated from each other in the vehicle vertical direction. The fourth upper wall **44a** is disposed on a relatively upper side, and the fourth lower wall **44b** is disposed on a relatively lower side. The fourth upper wall **44a** and the fourth lower wall **44b** have the same thickness. However, the thicknesses of the fourth upper wall **44a** and the fourth lower wall **44b** may be different.

(25) The fifth closed cross-sectional portion **45** includes a fifth upper wall **45a** and a fifth lower wall **45b** that extend in the vehicle width direction and are disposed to face each other while being separated from each other in the vehicle vertical direction. The fifth upper wall **45a** is disposed on a relatively upper side, and the fifth lower wall **45b** is disposed on a relatively lower side. The fifth upper wall and the fifth lower wall **45b** have the same thickness. However, the thicknesses of the fifth upper wall **45a** and the fifth lower wall **45b** may be different.

(26) The first to fifth closed cross-sectional portions **41** to **45** are partitioned by four partition walls **51** to **54** extending in the vehicle vertical direction. Specifically, the third closed cross-sectional portion **43** and the fourth closed cross-sectional portion **44** share the partition wall **51**. The fourth closed cross-sectional portion **44** and the fifth closed cross-sectional portion **45** share the partition wall **52**. The fifth closed cross-sectional portion **45** and the second closed cross-sectional portion **42** share the partition wall **53**. The second closed cross-sectional portion **42** and the first closed cross-sectional portion **41** share the partition wall **54**. The second closed cross-sectional portion **42** and the first closed cross-sectional portion **41** share the partition wall **54**. The four partition walls **51** to **54** are arranged at equal intervals in the vehicle width direction.

(27) An inner side wall **55** is disposed on the innermost side in the vehicle width direction of the reinforcing member **40**. The inner side wall **55** is included in the first closed cross-sectional portion **41** and extends in the vehicle vertical direction so as to connect the inner ends in the vehicle width direction of the first upper wall **41a** and the first lower wall **41b**. In addition, an outer side wall **56** is disposed on the outermost side in the vehicle width direction of the reinforcing member **40**. The outer side wall **56** is included in the third closed cross-sectional portion **43** and extends in the vehicle vertical direction so as to connect the outer ends in the vehicle width direction of the third upper wall **43a** and the third lower wall **43b**.

(28) Referring to FIG. 2, the first closed cross-sectional portion **41** is joined to the battery unit **10** by a bolt **60**. The bolt **60** extends in the vehicle vertical direction, penetrates the first upper wall **41a**, the first lower wall **41b**, the flange portion **13b** of the upper case **13**, the flange portion **14b** of the lower case **14**, and the floor panel **30**, and joins them. The bolt **60** is preferably inserted and fastened from bottom toward top from the viewpoint of efficiency of battery replacement. In addition, in order to secure high joint strength, it is preferable to suppress deformation of the fastening portion of the bolt **60** using the cylindrical collar **61** covering the periphery of the bolt **60**.

(29) In addition, the third closed cross-sectional portion **43** is joined to the side sill **20** by a bolt **70**. The bolt **70** extends in the vehicle vertical direction, penetrates the third upper wall **43a** and the third lower wall **43b**, and the outer member **21** of the side sill **20**, terminates in the internal space **S2**, and joins them. Similarly to the above description, the bolt **70** is preferably inserted and fastened from bottom toward top, and it is preferable to use a cylindrical collar **71** covering the periphery of the bolt **70**.

(30) Referring to FIG. 3, the thickness t_{u1} of the first upper wall **41a** is larger than the thickness t_{u2} of the second upper wall **42a** ($t_{u1} > t_{u2}$). In addition, the thickness t_{l1} of the first lower wall **41b** is larger than the thickness t_{l2} of the second lower wall **42b** ($t_{l1} > t_{l2}$).

(31) Preferably, the thickness t_{u1} of the first upper wall **41a** is 1.2 times or more the thickness t_{u2} of the second upper wall **42a** ($t_{u1} \geq 1.2 \times t_{u2}$). Preferably, the thickness t_{l1} of the first lower wall **41b** is 1.2 times or more the thickness t_{l2} of the second lower wall **42b** ($t_{l1} \geq 1.2 \times t_{l2}$). By specifically

defining the required thickness in this manner, deformation of the innermost closed cross-sectional portion can be prevented even in consideration of material variations and the like, and sufficient battery protection performance can be secured.

(32) The thickness $tu3$ of the third upper wall **43a** is larger than the thickness $tu4$ of the fourth upper wall **44a** ($tu3 > tu4$). The thickness $t13$ of the third lower wall **43b** is larger than the thickness $t14$ of the fourth lower wall **44b** ($t13 > t14$). Alternatively, the thickness $tu3$ of the third upper wall **43a** may be the same as the thickness $tu4$ of the fourth upper wall **44a**. In addition, the thickness $t13$ of the third lower wall **43b** may be the thickness $t14$ of the fourth lower wall **44b**.

(33) Preferably, the thickness $tu1$ of the first upper wall **41a** is smaller than the thickness $tu3$ of the third upper wall **43a** ($tu1 < tu3$). In addition, preferably, the thickness $t11$ of the first lower wall **41b** is smaller than the thickness $t13$ of the third lower wall **43b** ($t11 < t13$). By defining in this manner, the first upper wall **41a** and the first lower wall **41b** do not become excessively thick, so that it is possible to suppress an excessive increase in weight. In addition, in the extruded material made of an aluminum alloy, when a significant difference is provided in thickness, there is a possibility that extrusion processing cannot be stably executed. Therefore, by providing the upper limit to the thickness as described above, the extrusion processing can be stably executed.

(34) Any of the thicknesses $tu2$, $tu4$, and $tu5$ of the second, fourth, and fifth upper walls **42a**, **44a**, and **45a** is the same ($tu2 = tu4 = tu5$). In addition, any of the thicknesses $t12$, $t14$, and $t15$ of the second, fourth, and fifth lower walls **42b**, **44b**, and **45b** is the same ($t12 = t14 = t15$).

(35) The thicknesses $tp1$ to $tp4$ of the partition walls **51** to **54** gradually increase from the outer side toward the inner side in the vehicle width direction ($tp1 < tp2 < tp3 < tp4$). Accordingly, the degree of deformation can be equalized according to the progress of deformation of the reinforcing member **40** at the time of the side collision, and the energy absorption efficiency can be improved.

(36) According to the side sill reinforcement structure for an electric vehicle of the present embodiment, the following action and effect are produced.

(37) Since an extruded material made of an aluminum alloy is used as the reinforcing member **40**, the thickness can be partially changed. Accordingly, the thickness $tu1$ of the first upper wall **41a** can be made larger than the thickness $tu2$ of the second upper wall **42a** ($tu1 > tu2$), and the thickness $t11$ of the first lower wall **41b** can be made larger than the thickness $t12$ of the second lower wall **42b** ($t11 > t12$). In particular, since being disposed on the innermost side in the vehicle width direction in the closed cross-sectional portion group **40A**, the first closed cross-sectional portion **41** is a closed cross-sectional portion closest to the battery unit **10**. Furthermore, since extending in the vehicle width direction, the first upper wall **41a** and the first lower wall **41b** of the first closed cross-sectional portion **41** are portions that receive a side collision load. Therefore, increasing in the thickness of the portion efficiently improves battery protection performance while suppressing an increase in weight and an increase in cost as compared with a case of adopting a reinforcing member having a uniform thickness.

(38) In addition, since the first closed cross-sectional portion **41** having a relatively large thickness is firmly joined to the battery unit **10** by the bolt **60**, high coupling rigidity can be secured.

(39) In addition, the thickness $tu3$ of the third upper wall **43a** is made larger than the thickness $tu4$ of the fourth upper wall **44a** ($tu3 > tu4$), and the thickness $t13$ of the third lower wall **43b** is made larger than the thickness $t14$ of the fourth lower wall **44b** ($t13 > t14$). Accordingly, the strength of the outermost portion (third closed cross-sectional portion **43**) in the vehicle width direction first receiving deformation in the side collision is improved, and the peak value of the deformation load received at the initial stage of the side collision is increased. When the peak value of the deformation load received at the initial stage of deformation is increased as described above, the load variation according to the progress of deformation can be suppressed, and the energy absorption efficiency can be improved.

(40) In addition, since the first closed cross-sectional portion **41** having a relatively large thickness is firmly joined to the battery unit **10** by the bolt **60**, high coupling rigidity can be secured.

(41) In addition, since the third closed cross-sectional portion **43** having a relatively large thickness is firmly joined to the side sill **20** by the bolt **70**, high coupling rigidity can be secured.

Second Embodiment

(42) The reinforcing member **40** of the side sill reinforcement structure for an electric vehicle of the second embodiment shown in FIG. **4** is substantially the same as that of the first embodiment except for the thicknesses t_{u1} to t_{u5} of the first to fifth upper walls **41a** to **45a** and the thicknesses t_{l1} to t_{l5} of the first to fifth lower walls **41b** to **45b**. Therefore, the description of portions the same as those of the first embodiment may be omitted.

(43) The thicknesses t_{u1} to t_{u5} of the first to fifth upper walls **41a** to **45a** and the thicknesses t_{l1} to t_{l5} of the first to fifth lower walls **41b** to **45b** sequentially decrease from the outer side to the inner side in the vehicle width direction except for the first closed cross-sectional portion **41** (the first upper wall **41a** and the first lower wall **41b**). That is, the thicknesses t_{u2} to t_{u5} of the second to fifth upper walls **42a** to **45a** sequentially decrease from the outer side to the inner side in the vehicle width direction ($t_{u3} > t_{u4} > t_{u5} > t_{u2}$). In addition, the thicknesses t_{l2} to t_{l5} of the second to fifth lower walls **42b** to **45b** sequentially decrease from the outer side to the inner side in the vehicle width direction ($t_{l3} > t_{l4} > t_{l5} > t_{l2}$).

(44) In addition, the thickness t_{u1} of the first upper wall **41a** is smaller than the thickness t_{u3} of the third upper wall **43a** and larger than the thickness t_{u4} of the fourth upper wall **44a** ($t_{u4} < t_{u1} < t_{u3}$). The thickness t_{l1} of the first lower wall **41b** is smaller than the thickness t_{l3} of the third lower wall **43b** and larger than the thickness t_{l4} of the fourth lower wall **44b** ($t_{l4} < t_{l1} < t_{l3}$).

(45) According to the side sill reinforcement structure for an electric vehicle of the present embodiment, since the deformation strength of the reinforcing member **40** is sequentially lowered from the outer side to the inner side in the vehicle width direction in accordance with the deformation region being expanded in the vehicle front-rear direction according to the progress of the side collision, the load according to the progress of the deformation can be further equalized. Therefore, energy absorption efficiency can be further improved. In addition, since the application is excluded with respect to the first closed cross-sectional portion **41**, the first closed cross-sectional portion **41** can be configured less likely to be deformed even when the region on the outer side in the vehicle width direction from the first closed cross-sectional portion **41** is crushed. Therefore, higher battery protection performance can be secured.

Third Embodiment

(46) The side sill reinforcement structure for an electric vehicle of the third embodiment shown in FIG. **5** is substantially the same as that of the first embodiment except for the configuration related to the reinforcing member **40**. Therefore, the description of portions the same as those of the first embodiment may be omitted. It should be noted that in FIG. **5**, a portion indicated by a broken line circle is shown in an enlarged manner.

(47) In the present embodiment, the reinforcing member **40** has a lower side extending portion extending downward from the battery unit **10**. The lower side extending portion **40B** extends downward from the first closed cross-sectional portion **41**. In a cross section perpendicular to the vehicle front-rear direction, the lower side extending portion **40B** includes a sixth closed cross-sectional portion **46** having a rectangular shape.

(48) The lower side extending portion **40B** includes, so as to constitute the sixth closed cross-sectional portion **46**, an inner side wall **57** and an outer side wall **58** extending in the vehicle vertical direction, and a sixth upper wall **46a** and a sixth lower wall **46b** connecting the inner side wall **57** and the outer side wall **58** and extending in the vehicle width direction. It should be noted that the sixth upper wall **46a** of the sixth closed cross-sectional portion **46** and the first lower wall **41b** of the first closed cross-sectional portion **41** are the same portion. That is, the first closed cross-sectional portion **41** and the sixth closed cross-sectional portion **46** share the first lower wall **41b** (that is, the sixth upper wall **46a**). The sixth lower wall **46b** includes a flange portion **46c** extending inward in the vehicle width direction beyond the inner side wall **57**.

(49) A plurality of cross members **32** extending in the vehicle width direction are provided below the battery unit **10**. The cross member **32** is disposed in the vehicle width direction of the battery unit **10** along the lower surface of the battery unit **10**.

(50) The lower side extending portion **40B** is joined to the cross member **32** at the flange portion **46c**. The bolt **60** extends in the vehicle vertical direction, penetrates the sixth lower wall **46b**, the sixth upper wall **46a** (first lower wall **41b**), the first upper wall **41a**, the flange portion **13b** of the upper case **13**, the flange portion **14b** of the lower case **14**, and the floor panel **30**, and joins them.

(51) According to the side sill reinforcement structure for an electric vehicle of the present embodiment, when a side collision load is applied to the first closed cross-sectional portion **41**, the load can be transmitted to the cross member **32** through the lower side extending portion **40B**. Therefore, higher battery protection performance can be secured.

(52) A first modification of the third embodiment will be described with reference to FIG. **6**. It should be noted that in FIG. **6**, a portion indicated by a broken line circle is shown in an enlarged manner.

(53) In a cross section perpendicular to the vehicle front-rear direction, the shape of the sixth closed cross-sectional portion **46** of the lower side extending portion **40B** is not limited to a rectangle, and may be, for example, a trapezoid. In the example of FIG. **6**, the inner side wall **57** is disposed along the vehicle vertical direction, but the outer side wall **58** is disposed to be inclined from the vehicle vertical direction.

(54) In addition, the lower side extending portion **40B** does not need to be joined to the cross member **32** (see FIG. **5**). As shown in the example in FIG. **6**, when the battery unit **10** is sufficiently large in the vehicle vertical direction, the first closed cross-sectional portion **41** and the sixth closed cross-sectional portion **46** may be disposed to abut on the battery unit **10**. In this case, the deformation is suppressed by transmitting the side collision load to the bottom plate constituting the battery unit **10** rather than the cross member **32**.

(55) A second modification of the third embodiment will be described with reference to FIG. **7**.

(56) In the cross section perpendicular to the vehicle front-rear direction, the lower side extending portion **40B** (see FIGS. **5** and **6**) does not need to be provided in order to be joined to the cross member **32**. For example, the first closed cross-sectional portion **41** may be joined to the cross member **32**.

(57) In the example in FIG. **7**, the end portion **32a** on the outer side in the vehicle width direction of the cross member **32** is crushed. The end portion **32a** is joined to the first closed cross-sectional portion **41** by a bolt **60**. The bolt **60** extends in the vehicle vertical direction, penetrates the end portion **32a** of the cross member **32**, the first lower wall **41b**, the first upper wall **41a**, the flange portion **13b** of the upper case **13**, the flange portion **14b** of the lower case **14**, and the floor panel **30**, and joins them.

(58) In addition, although not shown in detail, when a cross member is disposed in the battery case **12**, the reinforcing member **40** may be disposed on an outer side in the vehicle width direction of the cross member so that a side collision load can be transmitted from the reinforcing member **40** to the cross member.

Fourth Embodiment

(59) The side sill reinforcement structure for an electric vehicle of the fourth embodiment shown in FIGS. **8** and **9** is substantially the same as that of the first embodiment except for the configuration related to the shape and arrangement of the reinforcing member **40** and the shape of the side sill. Therefore, the description of portions the same as those of the first embodiment may be omitted.

(60) In the present embodiment, the reinforcing member **40** is disposed on an inner side in the vehicle width direction of the side sill **20**, and is joined to the inner member **22** of the side sill **20** by a bolt **70**. The inner member **22** includes a step portion **22a** having a shape complementary to that of the reinforcing member **40**, and the reinforcing member **40** is disposed in the step portion **22a**.

(61) As described above, the arrangement of the reinforcing member **40** is not limited to below the side sill **20** as in the first to third embodiments, and may be on the inner side in the vehicle width direction of the side sill **20**. That is, the reinforcing member **40** may be disposed so as to connect the side sill **20** and the battery unit **10** in any mode.

(62) In addition, in the present embodiment, the closed cross-sectional portion group **40A** of the reinforcing member **40** includes three rectangular closed cross-sectional portions **41** to **43** (**44**). As compared with the first embodiment, the fifth closed cross-sectional portion **45** (see FIG. 2) is omitted, and the second closed cross-sectional portion **42** (see FIG. 2) and the fourth closed cross-sectional portion **44** (see FIG. 2) are integrated. Therefore, the closed cross-sectional portion group **40A** of the reinforcing member **40** includes a first closed cross-sectional portion **41** disposed on the innermost side in the vehicle width direction, a third closed cross-sectional portion **43** disposed on the outermost side in the vehicle width direction, and a second closed cross-sectional portion **42** (that is, the fourth closed cross-sectional portion **44**) disposed between the first closed cross-sectional portion **41** and the third closed cross-sectional portion **43**.

(63) The first to third closed cross-sectional portions **41** to **43** are partitioned by two partition walls **51** to **52** extending in the vehicle vertical direction. The third closed cross-sectional portion **43** and the second closed cross-sectional portion **42** share the partition wall **51**. The second closed cross-sectional portion **42** and the first closed cross-sectional portion **41** share the partition wall **52**.

(64) The first closed cross-sectional portion **41** is joined to the battery unit **10** by a bolt **60**. The bolt **60** extends in the vehicle vertical direction, penetrates the first upper wall **41a**, the first lower wall **41b**, the flange portion **13b** of the upper case **13**, and the flange portion **14b** of the lower case **14**, and joins them.

(65) As described above, the third closed cross-sectional portion **43** is joined to the inner member **22** of the side sill **20** by the bolt **70**. The bolt **70** extends in the vehicle vertical direction, penetrates the third upper wall **43a**, the third lower wall **43b**, and the step portion **22a** of the inner member **22**, terminates in the internal space **S2**, and joins them.

(66) The relative relationship between the thicknesses of the respective walls constituting the first closed cross-sectional portion **41**, the second closed cross-sectional portion **42** (that is, the fourth closed cross-sectional portion **44**), and the third closed cross-sectional portion **43** is the same as that of the first embodiment.

(67) As described above, the number of closed cross-sectional portions included in the reinforcing member **40** is not particularly limited, and may be 5 as in the first embodiment, may be 3 as in the fourth embodiment, or may be 4 or 6 or more.

(68) As described above, although the specific embodiments and their modifications of the present invention are described, the present invention is not limited to the above-described embodiments, and can be implemented with various modifications within the scope of the present invention. For example, an appropriate combination of contents of the individual embodiments and modifications may be one embodiment of the present invention.

(69) In addition, the shape of each closed cross-sectional portion of the reinforcing member **40** is not limited to a rectangle or a trapezoid as in each of the above embodiments, and may be any polygon (for example, a hexagon or the like) having an upper wall and a lower wall facing each other.

Claims

1. A side sill reinforcement structure for an electric vehicle comprising: a battery unit for drive disposed in a vehicle interior lower portion of an electric vehicle; a side sill extending in a vehicle front-rear direction on an outer side in a vehicle width direction of the battery unit; and a reinforcing member configured to connect the battery unit and the side sill, wherein the reinforcing member is an extruded material made of an aluminum alloy, and includes a closed cross-sectional

portion group in which a plurality of closed cross-sectional portions are continuous in the vehicle width direction in a cross section perpendicular to the vehicle front-rear direction, wherein the closed cross-sectional portion group includes a first closed cross-sectional portion disposed on an innermost side in the vehicle width direction and a second closed cross-sectional portion disposed adjacent to the first closed cross-sectional portion, wherein the first closed cross-sectional portion includes a first upper wall and a first lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in a vehicle vertical direction, wherein the second closed cross-sectional portion includes a second upper wall and a second lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein a thickness of the first upper wall is larger than a thickness of the second upper wall, and a thickness of the first lower wall is larger than a thickness of the second lower wall.

2. The side sill reinforcement structure for an electric vehicle according to claim 1, wherein a thickness of the first upper wall is 1.2 times or more a thickness of the second upper wall, and a thickness of the first lower wall is 1.2 times or more a thickness of the second lower wall.

3. The side sill reinforcement structure for an electric vehicle according to claim 1, wherein the first closed cross-sectional portion is joined to the battery unit by a bolt.

4. The side sill reinforcement structure for an electric vehicle according to claim 1, wherein the closed cross-sectional portion group includes a third closed cross-sectional portion disposed on an outermost side in the vehicle width direction and a fourth closed cross-sectional portion disposed on an inner side in the vehicle width direction adjacent to the third closed cross-sectional portion, wherein the third closed cross-sectional portion includes a third upper wall and a third lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, wherein the fourth closed cross-sectional portion includes a fourth upper wall and a fourth lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein a thickness of the third upper wall is larger than a thickness of the fourth upper wall, and a thickness of the third lower wall is larger than a thickness of the fourth lower wall.

5. The side sill reinforcement structure for an electric vehicle according to claim 4, wherein a thickness of the first upper wall is smaller than a thickness of the third upper wall, and a thickness of the first lower wall is smaller than a thickness of the third lower wall.

6. The side sill reinforcement structure for an electric vehicle according to claim 4, wherein the third closed cross-sectional portion is joined to the side sill by a bolt.

7. The side sill reinforcement structure for an electric vehicle according to claim 1, wherein each closed cross-sectional portion included in the closed cross-sectional portion group includes an upper wall and a lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein thicknesses of the upper wall and the lower wall are sequentially decreased from an outer side to an inner side in the vehicle width direction excluding the first closed cross-sectional portion.

8. The side sill reinforcement structure for an electric vehicle according to claim 1, wherein the reinforcing member includes a lower side extending portion extending downwards the battery unit, wherein a cross member extending in the vehicle width direction is provided under the battery unit, and wherein the lower side extending portion is joined to the cross member.

9. The side sill reinforcement structure for an electric vehicle according to claim 2, wherein the first closed cross-sectional portion is joined to the battery unit by a bolt.

10. The side sill reinforcement structure for an electric vehicle according to claim 2, wherein the closed cross-sectional portion group includes a third closed cross-sectional portion disposed on an outermost side in the vehicle width direction and a fourth closed cross-sectional portion disposed on an inner side in the vehicle width direction adjacent to the third closed cross-sectional portion,

wherein the third closed cross-sectional portion includes a third upper wall and a third lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, wherein the fourth closed cross-sectional portion includes a fourth upper wall and a fourth lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein a thickness of the third upper wall is larger than a thickness of the fourth upper wall, and a thickness of the third lower wall is larger than a thickness of the fourth lower wall.

11. The side sill reinforcement structure for an electric vehicle according to claim 10, wherein a thickness of the first upper wall is smaller than a thickness of the third upper wall, and a thickness of the first lower wall is smaller than a thickness of the third lower wall.

12. The side sill reinforcement structure for an electric vehicle according to claim 10, wherein the third closed cross-sectional portion is joined to the side sill by a bolt.

13. The side sill reinforcement structure for an electric vehicle according to claim 2, wherein each closed cross-sectional portion included in the closed cross-sectional portion group includes an upper wall and a lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein thicknesses of the upper wall and the lower wall are sequentially decreased from an outer side to an inner side in the vehicle width direction excluding the first closed cross-sectional portion.

14. The side sill reinforcement structure for an electric vehicle according to claim 10, wherein each closed cross-sectional portion included in the closed cross-sectional portion group includes an upper wall and a lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein thicknesses of the upper wall and the lower wall are sequentially decreased from an outer side to an inner side in the vehicle width direction excluding the first closed cross-sectional portion.

15. The side sill reinforcement structure for an electric vehicle according to claim 11, wherein each closed cross-sectional portion included in the closed cross-sectional portion group includes an upper wall and a lower wall that extend in the vehicle width direction and are arranged to face each other while being separated from each other in the vehicle vertical direction, and wherein thicknesses of the upper wall and the lower wall are sequentially decreased from an outer side to an inner side in the vehicle width direction excluding the first closed cross-sectional portion.

16. The side sill reinforcement structure for an electric vehicle according to claim 2, wherein the reinforcing member includes a lower side extending portion extending downwards the battery unit, wherein a cross member extending in the vehicle width direction is provided under the battery unit, and wherein the lower side extending portion is joined to the cross member.
